CRISM Hyperspectral Pipeline

A Step-by-Step Scientific & Technical Methodology

Target Sensor: MRO CRISM (L & S Sensors)
Date: July 2026
Status: Stages 1–4 Completed
Output Format: Calibrated & Denoised Cubes (*_S4.npy)

Executive Summary

The Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) on board the Mars Reconnaissance Orbiter (MRO) collects hyperspectral images of the Martian surface in hundreds of infrared and visible wavelengths. These signals allow scientists to identify minerals that formed in water (like clays and salts), helping reconstruct Mars' wet past. However, the raw data is heavily corrupted by spacecraft heat, camera electronics anomalies, cosmic rays, and atmospheric dust.

This pipeline functions as a multi-stage correction facility. It systematically separates physical phenomena (such as temperatures and atmospheric paths) from camera electrical anomalies (such as stripes and random static). By applying physical laws first and advanced self-supervised machine learning next, the pipeline cleans the data while guaranteeing that real mineral markers are never altered or removed.

Pipeline Stages at a Glance

Stage	Name	Core Action (Simple English)	Primary Targets
Stage 1	Exploratory Data Analysis (EDA)	Data inspection, sanity checks, and marking broken channels.	Dead, saturated, and bad spectral bands.
Stage 2	Noise Characterisation	Sorting the noise into different categories based on their physical source.	Stripes, random static, spikes, thermal heat, saturation.
Stage 3	Physics-Based Corrections	Correcting issues that follow known physical laws (like heat and shadows).	Cosmic spikes, satellite shadows, thermal background, atmospheric dust.
Stage 4	ML Denoising	Using self-supervised AI models to clean complex electronics noise.	Camera readout stripes, random sensor background static.

Stage 1: Exploratory Data Analysis (EDA)

Simple Explanation: Testing the data's health and marking broken sensor parts.

Before cleaning, the pipeline opens the CRISM data files and checks their integrity. It verifies that the reflectance values (measured as I/F) are within physical limits (between a dark current floor of -0.05 and a bright surface ceiling of 1.3). More importantly, it catalogs the health of every single spectral band:

- **Dead Bands:** Sensor channels that are completely broken and return zero signal (variance < 10^{-12}).
- **Noisy Bands:** Channels displaying extreme signal fluctuations (standard deviation in the top 5%).
- **Low Signal Bands:** Channels where light levels drop too low to provide reliable data (mean < 10^{-4}).

Why it matters: Modern space sensors consist of hundreds of miniature light detectors. Over time, some deteriorate due to cosmic radiation. If we do not identify and mask these broken channels, they will contaminate all our mathematical calculations and machine learning steps later.

Stage 2: Noise Characterisation

Simple Explanation: Sorting the sensor noise into different bins based on their origin.

Different types of noise require different solutions. In this stage, the pipeline analyzes the image's mathematical properties to map out five distinct types of errors:

- **N1 Column Stripes:** Vertical stripes caused by differences in how individual detector columns read out signal. We verify this by checking if the stripe pattern is highly correlated across different wavelengths.
- **N2 Gaussian Random Noise:** Standard background static, similar to TV fuzz. We flag bands where the signal is less than 5 to 15 times stronger than the static (SNR < 15).
- **N3 Spikes:** Single, extremely bright pixels caused by cosmic rays hitting the sensor. Detected using a robust statistical threshold (6 times the Median Absolute Deviation).
- **N4 Thermal Excess:** Background heat radiating from Mars and the spacecraft. It behaves like a thermal shroud, masking mineral signatures at wavelengths longer than 3.0 microns.
- **N5 Saturation:** Blinding of the sensor from extremely bright reflections, capping the data at full capacity.

Why it matters: A physician must diagnose an illness before prescribing medicine. Similarly, the pipeline must know if a sensor error is a spike, a stripe, or heat to apply the correct mathematical cure.

Stage 3: Physics-Based Corrections

Simple Explanation: Fixing errors using standard formulas derived from the laws of physics.

This stage corrects errors that follow predictable physical processes. It does not use AI; instead, it relies on physical geometry and thermodynamic models:

Illumination:	Equalizes light levels across the image. It divides each row by its median brightness to remove shadowing caused by local topography and spacecraft viewing angles.
Spikes (N3):	Finds single-pixel cosmic ray hits and replaces them with the average value of the surrounding 8 pixels in a 3x3 spatial window.
Saturation (N5):	Fills in blinded, saturated pixels by calculating the average of the immediate neighboring wavelengths (spectral interpolation).
Thermal (N4):	Fits a Planck blackbody curve (representing thermal heat radiation) to the long-wavelength infrared region and subtracts this heat signature from every pixel's spectrum.
Atmosphere:	Adjusts the spectrum's overall tilt by comparing it to a known dark surface at 1.0 and 2.5 microns, correcting the broad slope caused by suspended atmospheric dust.

Scientific Review Note: *A minor discrepancy was identified in the N4 Thermal correction. The model fits the Planck blackbody curve directly to I/F (reflectance) data. Physically, reflectance is normalized by incoming solar flux, which decreases in the infrared. To make this physically rigorous, the Planck curve must be scaled by the solar flux model before subtraction. Additionally, narrow atmospheric gas bands (like the 2.0-micron CO2 band) are currently masked rather than physically modeled.*

Stage 4: Machine Learning Denoising

Simple Explanation: Using self-supervised AI models to clean complex electronics noise.

Some noise patterns, like random static (N_2) and fine camera column stripes (N_1), are too complex to fix with simple formulas. In Stage 4, the pipeline trains two separate AI models on the image itself, without needing pre-existing clean examples:

1. Stripe CNN (N1): A 1D spatial Convolutional Neural Network. The model first applies a basic column-subtraction algorithm, then uses a deep convolutional network to study the remaining subtle stripes and clean them up.

2. Spectral DAE (N2): A Denoising Autoencoder trained via the *Noise2Void* method. We take the spectrum of a pixel (its light curve across all colors) and randomly hide 10% of the bands. We force the AI to reconstruct the missing values from adjacent wavelengths. Because random noise is independent from band to band, the AI cannot guess it. It is forced to learn the smooth, underlying mineral signatures.

Scientific Safeguards: The AI is heavily restricted to prevent it from altering real geological details:

- **Smoothness Loss:** The model is penalized if the noise it extracts has sharp spectral features. This forces the model to keep sharp absorption bands (which indicate minerals) in the clean output.
- **Automatic Validation:** The pipeline measures the spectral angle (SAM) and key mineral indexes (like OLINDEX3 for olivine) before and after. If the AI shifts mineral index values by more than 20%, it enters a *REVIEW* state and automatically blends 30% of the raw physical data back in. If it fails, the AI output is discarded entirely.

Validation Results Summary

In test runs, Stage 4 achieved significant improvements in clarity without corrupting diagnostic features. For example, the L-sensor scene **frt0001458f_01_if156l** saw a Signal-to-Noise Ratio (SNR) increase from **21.03 to 29.45**, while the olivine feature (OLINDEX3) was preserved within a tiny margin of 2.8% from its original value. All processed files were successfully saved as standardized NumPy arrays (*_S4.npy).